ABRAR JAWAD TARAFDER

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Education

York University, Lassonde School of Engineering **Toronto, ON**
Honours Bachelor of Science, Computer Science *Apr 2022 – Aug 2026*

- **GPA:** 3.8/4.0 - Graduated with First Class Standing
- **Relevant Coursework:** Advanced OOP, Data Structures, Algorithm Design, Advanced Data Structures & Algorithms, Discrete Mathematics I & II, Calculus I, II & III, Linear Algebra, Probability, Theory of Computer Science, Software Design, Operating Systems, Database Management Systems, Data Mining, Computer Vision, Information Networks

Ontario Tech University **Oshawa, ON**
Honours Bachelor of Science, Computer Science (Transferred to York University) *Sep 2021 – Apr 2022*

Experience

York University **May 2026 – Aug 2026**
Database Developer & Research Assistant *Toronto, ON, Canada*

- Developed and maintained **relational databases** using **SQL** and **Python**, building pipelines to clean, merge, and standardize **10,000+ historical records**. Automated SQL workflows, reconciling **10,000+ record IDs** in **minutes**.
- Automated the extraction of historical personnel records from **28 archival PDF documents** using **Python**, integrating **18+ new role definitions** into the research database.

Hydro One **May 2024 - Aug 2025**
OT Transmissions Application Intern, Co-op *Barrie, ON, Canada*

- Developed an in-house application for biweekly database releases using ReactJS, Bootstrap, and PostgreSQL, ensuring responsive design and **optimized server performance**.
- **Automated error handling** in GitLab by building Python scripts to parse and convert logs into XML, improving external pipeline workflows.
- Developed reusable GitLab **CI/CD pipelines** with PERL and PowerShell, automating manual workflows and achieving a **98%** efficiency gain while improving scalability and consistency.
- Created a general shuttle pipeline template to accelerate server-to-server file transfers, reducing time by **73%**
- Automated the GE EMS LiveSnap loading process by developing GitLab pipeline and PERL scripts, **reducing load time** by **87%**.
- **Maintained QA and Development** servers in GE EMS, performing efficient monthly updates of savecases and equipment data.
- Performed data uploads and mapping for asset acquisition, ensuring nameplate tracking and in-service documentation.
- Developed SQL validation scripts within GE Source to automate field checks, triggering **real-time error alerts** and improving entry accuracy for the Application Modelling team; version-controlled in GitLab.

WalkWithWeb Inc. **May 2023 – May 2024**
Web Developer & Research Assistant *Toronto, ON, Canada*

- Researched, designed, and developed websites using HTML, CSS, Bootstrap, and JavaScript, facilitating seamless user interaction within a visually captivating environment amongst **3 departments**.
- Crafted Figma **interactive wireframes** for the websites and contributed significantly to marketing strategies, social media analysis, and graphic design, increasing the amount of output by **50%**.

Leadership Experience

Jarvis Consulting Group **Apr 2023 - Apr 2024**
Student Ambassador Team Lead *Toronto, ON, Canada*

- Leading a team of **14 Computer Science and Engineering** students to engage in immersive tech-related events.

Projects

Cruise Critic SQL Power BI
- Built a **Power BI** star schema using **SQL and DAX**, analyzing cruise revenue, profitability, and destination demand.

TTC Network Analysis (Group Project)
- Analyzed **9,406 TTC nodes** using **Python, NetworkX, Dijkstra, A*, and PageRank** to evaluate transit routing.

Freshroute Full-Stack Grocery Platform (Group Project)
- Built a full-stack grocery app using **React, Node.js, Express.js, and MongoDB** with REST APIs and checkout.

Atlas AI Research Workspace
- Developed an **AI-powered research agent** using **Python, LangChain, OpenAI APIs, and Streamlit**, integrating **web search, tool calling, conversational memory, and structured responses** with PDF export.

Technical Skills

Languages: Python, Java, JavaScript, C++, SQL, PHP, HTML/CSS, PowerShell, Perl, R | **Frameworks:** React, Node.js, Express.js, LangChain, Streamlit, Bootstrap, NetworkX, TKinter | **Data:** SQL, MongoDB, PowerBI, DAX, Excel | **Tools:** Git, GitLab CI/CD, PostgreSQL, MongoDB, Power BI, DAX, Tableau, OpenAI API, REST APIs, Power Automate, Jira, Figma